

Milestone 3 Completion Report

Project Title: BankBot - An Chatbot for Banking FAQs (Brand: "Bank of Trust")

Milestone Title: Milestone 03 - User Interface (UI) Implementation & Full-Stack Integration

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1. Executive Summary

This report documents the successful completion of Milestone 3 for the **"BankBot - An Chatbot for Banking FAQs"** project, formally branded as **Bank of Trust**. The primary objective of this phase was to transition from backend logic to a fully functional, user-facing application. By leveraging the **Streamlit** framework, a responsive and visually professional frontend was developed. This interface has been successfully integrated with the **spaCy** NLU backend, allowing for a seamless "vertical slice" of functionality where users can log in, view financial data, and interact with the AI assistant in real-time.

2. Technical Implementation

The development architecture for Milestone 3 focused on modularity and user experience. The following technical strategies were employed:

- **Frontend Framework (Streamlit):** Python's Streamlit library was selected for its ability to rapidly deploy data-driven web applications. It serves as the presentation layer, rendering dynamic content based on user interaction.
- **Session State Management:** To maintain context across page re-runs—a necessity for a stateless web framework—`st.session_state` was utilized extensively. This handles persistence for user authentication status (Login/Logout), navigation flags, and the appending of chat logs during the conversational flow.
- **Mock Logic Architecture:** To enable "vertical slice" testing before the final database integration (Milestone 4), a mock data layer was constructed. This simulates API responses for user profiles, account balances, and transaction histories, allowing the UI components to be rigorously tested against realistic data structures.
- **Custom Styling (CSS):** Standard Streamlit components were enhanced using injected CSS. A specific color palette comprising **Deep Navy** backgrounds/sidebars) and **Emerald Green/Electric Blue** (accents/buttons) was applied to align with the "Nexus Prime" corporate identity, ensuring the application resembles a commercial FinTech product rather than a standard data dashboard.

3. Key Deliverables

The following distinct functional modules were delivered and verified during this phase:

A. Secure Authentication & Landing Page

The application features a secure entry point simulating a real-world banking portal. The landing page provides options for "Retail" and "Corporate" logins. The login interface utilizes form masking for password security and verifies credentials against the mock user directory. Upon successful validation, the session state is updated to grant access to private routes.

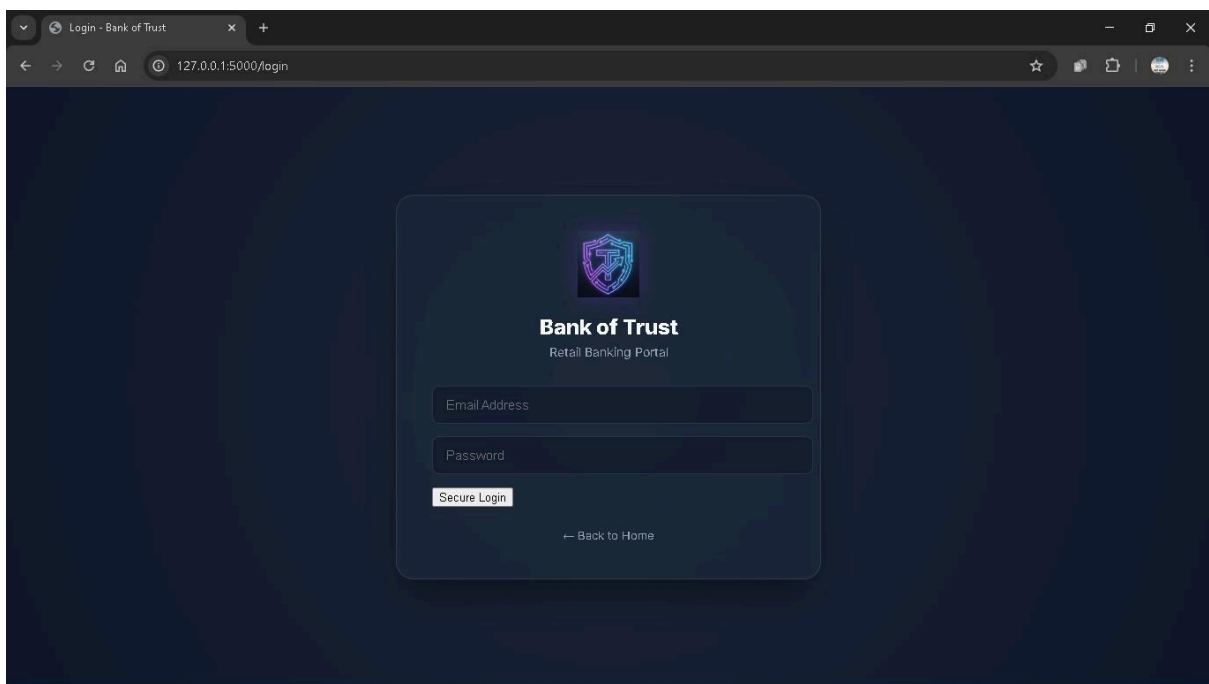


Figure 1: The Secure Login Interface demonstrating the Nexus Prime branding and input validation.

B. Customer Dashboard

Post-authentication, the user is directed to a comprehensive dashboard. This view dynamically renders:

- **Account Overview:** A prominent display of the Total Balance (e.g., ₹320,000).
- **User Details:** Dynamic fetching of user metadata, including Branch Details (Hyderabad Main) and Account Number.
- **Transaction History:** A tabular view of recent financial activities, displaying dates, descriptions (e.g., "Sent to Imran"), payment modes (UPI/Bank Transfer), and status flags.

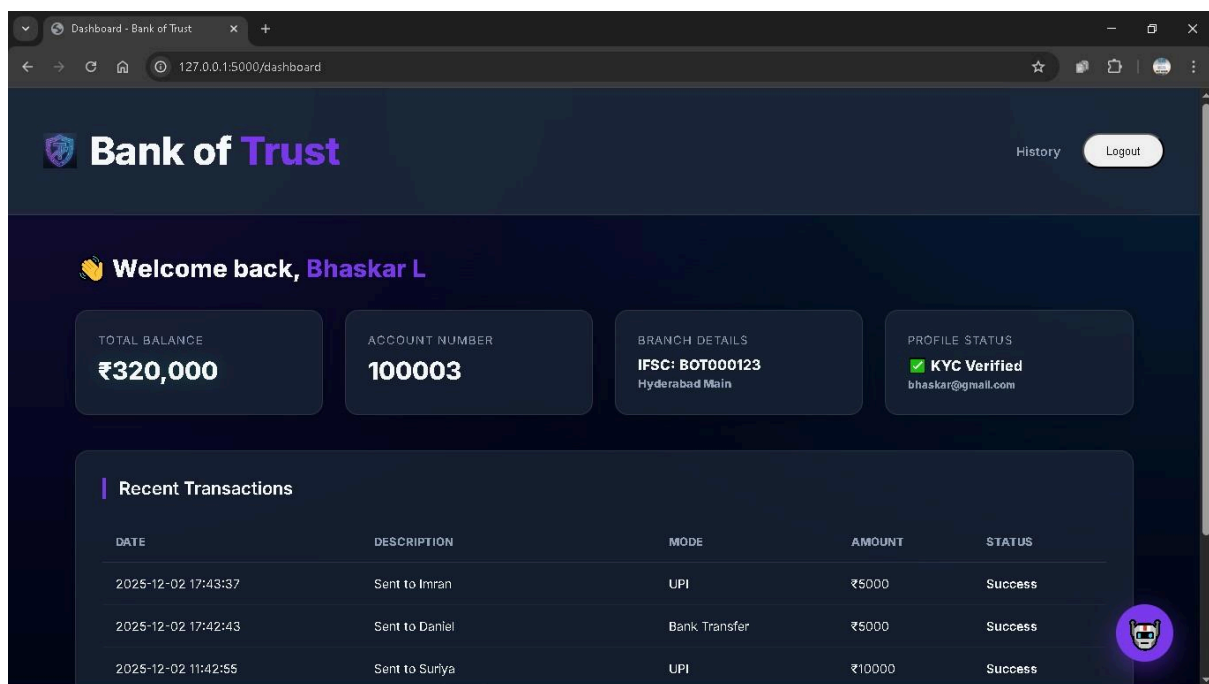


Figure 2: The Main Dashboard displaying real-time balance and transaction history for the logged-in user.

C. Conversational Interface (AI Assistant)

The core feature of the project, the AI Chatbot, is embedded directly into the application. It features a modern chat interface with distinct styling for user queries (aligned right) and bot responses (aligned left). The interface connects directly to the spaCy backend to process intents. As evidenced in the testing logs, the bot can successfully parse intents such as `balance_enquiry` and retrieve specific data (e.g., "Account 100003: Available Balance is ₹320,000") without the user leaving the chat window.

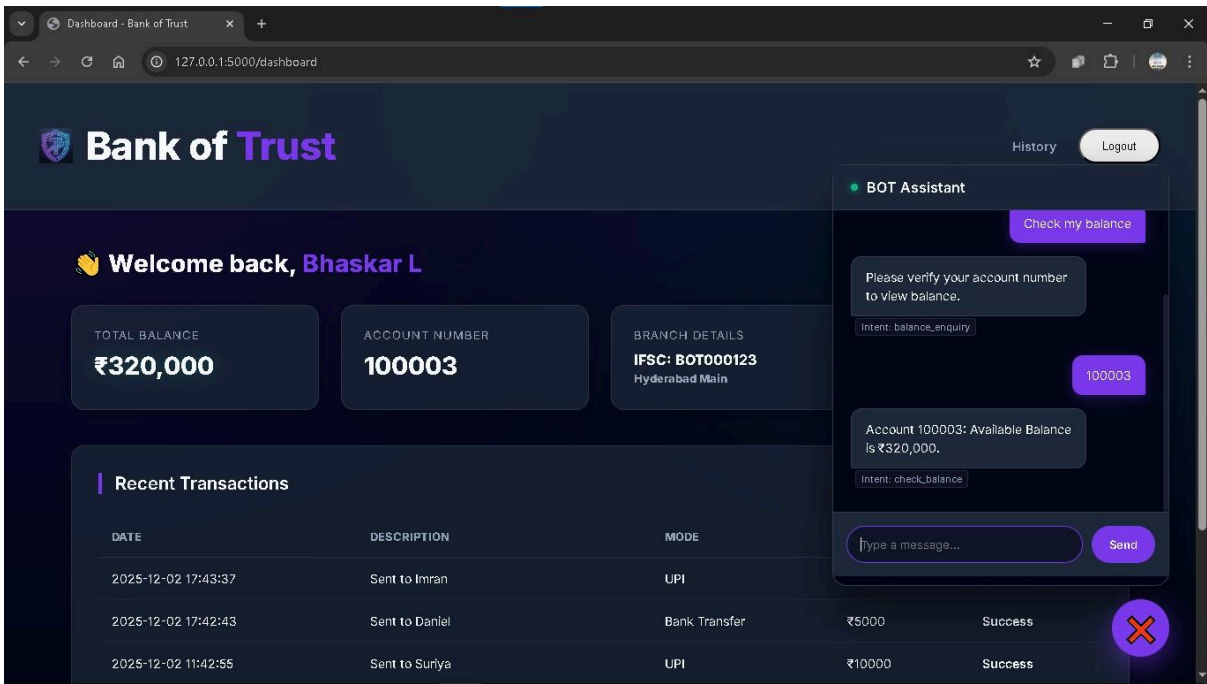


Figure 3: The AI Assistant overlay demonstrating intent recognition (Balance Check) and context-aware responses.

4. Challenges & Solutions

Challenge: Integrating Custom CSS with Streamlit Streamlit is designed primarily for standard data layouts, making custom branding challenging. Applying the specific "Nexus Prime" Deep Navy theme required overriding default iframe styles.

- **Solution:** I implemented a custom CSS injection module using `st.markdown(unsafe_allow_html=True)`. This allowed for the modification of the main container background, sidebar styling, and the creation of custom "chat bubbles" that visually distinguish the AI from the user.

Challenge: State Persistence in Chat Initially, the chat history would clear every time the user interacted with a different UI element (like the "History" tab).

- **Solution:** I restructured the `chat_history` list to be initialized only once within `st.session_state`. This ensures that the conversation context is preserved throughout the user's active session, even as they navigate between the dashboard and logs.

5. Conclusion & Next Steps

Milestone 3 has successfully delivered a stable, aesthetically professional, and functional User Interface. The integration between the Streamlit frontend and the spaCy NLU backend is operating with low latency, and the mock logic successfully demonstrates the intended user journey.

Next Steps (Milestone 4):

- Replace Mock Logic with live database connections (SQL/MongoDB).
- Implement final transactional logic (updating balances in real-time).
- Conduct end-to-end user acceptance testing.

The project is currently on schedule and meets all technical requirements set forth in the internship roadmap.